Support for immigration reduction and physician distrust in the United States Replication

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Abstract

This study aims to replicate the findings of the 2016 paper "Support for immigration reduction and physician distrust in the United States" by Frank L Samson. Samson's paper utilizes data from the 2012 General Social Survey (GSS) to examine whether physician distrust is tied to attitudes around immigration. The analysis uses a sample of 1,080 responses from the GSS. The replication follows the methodology of the original study, using a five-item physician distrust scale and a single-item measure, a single item measure of immigration attitude, and control demographic, political, and health-related variables. The study utilizes ordinary least squares (OLS) regression for the multi-item scale and ordinal logistic regression (OLR) for the single-item measure, with multiple imputation to handle missing data. Descriptive statistics closely matched the original study, and regression results largely replicated the positive association between immigration reduction support and physician distrust. Minor differences in coefficients and significance levels are likely due to multiple imputation and coding variations in control variables. Overall, the replication supports the main conclusion that support for immigration reduction is positively linked with physician distrust, suggesting physician distrust could also be tied to general declining social trust in the United States.

Introduction

Frank L Samson's 2016 article "Support for immigration reduction and physician distrust in the United States" evaluates the correlation between attitudes about immigration and distrust of physicians. Prior research shows that trust in doctors has decreased over the past 50 years. At the same time, there has also been a decrease in general social trust, especially in attitudes surrounding immigration. This study explores whether individuals who support reducing immigration are more likely to report higher levels of physician distrust, indicating that physician distrust may be tied to general social attitudes rather than just physician-patient interactions.

To conduct this investigation, Samson utilized data from the 2012 General Social Survey (GSS). The sample size includes 1,080 respondents Physician distrust is measured using a five-item scale assessing general trust, medical skills, treatment option transparency, physician priorities, and admitting mistakes, as well as a single-item measure focused solely on general trust in physicians. Support for immigration reduction is measured on a five-point scale asking whether immigration should be increased or decreased. The analysis also includes many control variables including age, education, family income, gender, race, political views, foreign-born status, employment status, type of health insurance, and self-rated health status.

Samson uses ordinary least squares regression with the multiple item scale and ordinal logistic regression with the single-item measure, as well as utilizing multiple imputation to handle missing data. The results show a significant positive relationship between support for reducing immigration and physician distrust with and without controlling for demographic, political, and health-related factors. Samson's findings suggest that immigration attitude can be used as a predictor for physician distrust, indicating that physician distrust is tied to general social climate rather than solely by individual experiences.

This replication study aims to reproduce Samson's main findings using the same 2012 General Social Survey data and methodologies used in the source article to verify the relationship between immigration attitudes and physician distrust.

Data and Methods

This replication study uses the same dataset as the source article, the General Social Survey (GSS) from 2012. From this dataset, I selected 16 variables for analysis including five variables to measure physician distrust, one to measure immigration attitude, and 10 control variables to account for demographic and health related factors, chosen according to the measures section on page 3 of the source article. The variables chosen were: doctrst, docearn, docskls, doctlk, docmstk, LETIN1, age, educ, realinc, sex, race, polviews, born, wrkstat, instype, and HEALTH1. The original study had a sample size of 1,080, and I was able to match that exactly by filtering out respondents who weren't asked one or more of the questions. {*mmarks9-PDIR-01-mgmt-setup*}

Data management and cleaning involved several steps. First, I created new versions of two of the physician distrust variables (docearn and docskls) and reverse coded them to match the original study {*mmarks9-PDIR-02-mgmt-revcode*} Then, I created new versions of each variable that contained missing data or undecided responses ("Can't choose"/ "Don't know"/ "No answer") and coded those responses as NA {*mmarks9-PDIR-03-mgmt-missval*} Before taking the descriptive statistics needed to replicate Table 1, I also created a new version of the sex variable and recoded it to be binary, as well as editing the family income variable to recode some of the very low values as missing to match the percent of missing values in the source paper. {*mmarks9-PDIR-04-mgmt-table1var*}

To create the descriptive statistics for Table 1, I used the original versions of the physician distrust, immigration attitudes, political views, health status, and race variables to get proportions for each response option (e.g., the five-point immigration question) so that "Can't choose," "Don't know," and "No answer" responses were included. I didn't use the reverse-coded versions of docearn and docskls here, since reverse coding is only needed for the regression models. For other variables, like income, education, age, sex, foreign-born status, employment, and health insurance, I used the updated versions to calculate means, standard deviations, single proportions (e.g., percent of respondents who were unemployed), or percent missing values. {*mmarks9-PDIR-05-analysis-desc*}

To prepare for regression analysis, missing data was filled in via multiple imputation using the *mice* package {*mmarks9-PDIR-06-mgmt-impute*}. To create the multi-item physician distrust scale, I standardized and then averaged the five doctor variables. {*mmarks9-PDIR-07-mgmt-scale*}. Then, I took the average, standard deviation, and range of the scale to be included in Table 1. {*mmarks9-PDIR-08-analysis-table1scale*}

The analysis replicated both the regression approaches used in the original study. I used ordinary least squares (OLS) regression with the standardized multi-item physician distrust scale, {*mmarks9-PDIR-09-analysis-OLSreg*} and ordinal logistic regression (OLR) with the single-item physician distrust measure, utilizing the *ordinal* package. {*mmarks9-PDIR-10-analysis-OLRreg*}

Results and Discussion

The descriptive statistics in this replication closely match those reported in the original study, indicating that the sample was successfully recreated. Most measures included in Table 1 were very close to those found in the source article. The mean education in the replication was 13.8 years (SD = 3.0) versus 13.7 years (SD = 2.9) in the original study. Age averaged 50.1 years in the replication compared to 48.0 years in the original, with identical standard deviations of

17.1 years. {*mmarks9-PDIR-05-analysis-desc*} The physician distrust scale had a mean of 0.00 (SD = 0.68) with a range from −1.9 to 2.0, nearly identical to the original mean of −0.002 (SD = 0.67). {*mmarks9-PDIR-08-analysis-table1scale*} (Samson, 4)

Support for immigration reduction had some minor differences: the “Reduced a little” category was 26.3% in the replication versus 27.3% in the original, and “Reduced a lot” was 23.9% versus 22.8%. (Samson, 4) Items on the physician distrust scale (e.g., trust, discuss, skills, earnings, mistake) were all within a few percentage points of the original values. {*mmarks9-PDIR-05-analysis-desc*}.

A few variables showed slightly larger deviations. Foreign-born respondents were 9.5% in the replication versus 12.6% in the original, and the proportion with no health insurance was 15.6% versus 18.2%. Race distributions were nearly identical for Black respondents (15.3% vs. 15.1%), but the White and Other categories differed by approximately 3%. Family income was slightly lower in the replication with a mean of 36.8 (thousand) vs. 38.7 (thousand) and minor differences in standard deviation and percent missing. (Samson, 4) While these statistics are different from the original study, they are not very likely to impact the results of the regression significantly because they are used as controls and the differences are not major. {*mmarks9-PDIR-05-analysis-desc*}.

Models 1-3 use the multi-item physician distrust scale and OLS regression to look at the association between physician distrust and immigration attitudes. The results from Models 1–3 in Table 2 also closely align with those reported in the original study.

In Model 1, support for immigration reduction is positively and significantly associated with physician distrust (coefficient= 0.077, $p < .001$), matching the direction and significance of the original finding, though with a slightly smaller coefficient. {*mmarks9-PDIR-09-analysis-OLSreg*} (Samson, 5)

Model 2 adds the demographic control variables (age, education, etc.). Again, the coefficient for immigration attitude matches the original in direction (positive) and level of significance ($p < .001$), but the coefficient is slightly smaller in the replication. Also in the source article, the income variable is significant at $p<0.05$. (Samson, 5) In the replication, the p-value is 0.050055, so it is extremely close to the same level of significance. {*mmarks9-PDIR-09-analysis-OLSreg*}

Model 3 adds political views, employment, and health variables. The immigration coefficient remains positive, but its level of significance is slightly weaker in the replication ($p = 0.0012$) compared to the original study ($p < .001$). (Samson, 5)While the significance levels do not match exactly, they are very close. Health status shows the same level of significance in both models, with $p < .001$ in the original and $p = 3.99e{-}05$ in the replication. (Samson, 5){*mmarks9-PDIR-09-analysis-OLSreg*}

Model 4 uses the single-item physician distrust measure, ordinal logistic regression (OLR) and includes all variables. Compared to the original study, the replication shows some substantial differences. In this replication, support for immigration reduction is positively and significantly associated with physician distrust (0.175, $p < .01$), consistent in direction with the original study, though the original coefficient was larger and had a higher level of significance (0.265, $p < .001$). (Samson, 5)This variable is the main focus of the model because this study aims to examine how attitudes toward immigration relate to physician distrust, so it is important that values in the replication are consistent with the original. {*mmarks9-PDIR-10-analysis-OLRreg*}

Several control variables differ from the original study in significance. Age is not significant in the replication (p = 0.13), whereas in the original it was significant (0.01 < p < 0.05). (Samson, 5) Education and sex also show very strong significance in the replication (p = 0.0044 and 0.00547, respectively) while they are not significant at all in the source article. {*mmarks9-PDIR-10-analysis-OLRreg*}

The race variable for Black respondents is significant in both studies, though at different levels, The p-value in the replication (0.000556) is lower than the original study (0.001 < p < 0.01). Health status has similar significance in both models, with p-values between 0.01 and 0.05. (Samson, 5){*mmarks9-PDIR-10-analysis-OLRreg*}

Some of the differences in this model might come from using multiple imputation in the replication, which adds a small element of randomness, or from small differences in how variables like race and income were coded. The original study also omitted the intercept, so they could have ran the OLR without an intercept or just omitted it afterwards, Samson did not indicate either way. I ran the replication with the intercept, then omitted it afterwards. Even with differences in the significance of control variables, the positive link between support for immigration reduction and physician distrust matches what the original study found, which is the main focus of the model and the study.

Conclusion

This replication study supports the findings of Samson (2016), showing a consistent significant positive association between physician distrust and support for immigration reduction.

The descriptive statistics in Table 1 closely match the original study demonstrating that the sample was successfully recreated. Minor differences like slightly lower percentages of foreign-born respondents and health insurance coverage, and small variations in family income and race distributions show that there is some variation between the original and replication samples, but these variables are used as controls and are not very likely to impact the results of the regression significantly.

Models 1–3 closely mirror the original study's findings. Support for immigration reduction consistently shows a positive and significant association with physician distrust. While there are some small deviations in coefficients and p-values (e.g., the income variable in Model 2 and immigration coefficient in Model 3), these do not change the interpretation of the model.

Model 4 shows a few bigger differences from the original study. Support for immigration reduction is still positively associated with physician distrust, but the coefficient is a bit smaller and the significance slightly weaker than in Samson's results. There are also some major differences in control variables, like age, education, and sex. Despite these differences, there is still a consistent significant positive correlation between the immigration and physician distrust variables between the studies.

Overall, this replication supports Samson's conclusion that physician distrust and attitudes surrounding immigration are linked, and physician distrust may be linked to declining general social trust rather than just trust in healthcare.

Bibliography

Samson, Frank L. “Support for immigration reduction and physician distrust in the United States.” *SAGE open medicine* vol. 4 2050312116652567. 21 Jun. 2016, doi:10.1177/2050312116652567